

Emerging Wireless Standards for Next-Gen IoT Applications

This article is based on a talk by Rittu Sachdev, RF engineer, Texas Instruments, at India Electronics Week 2019 organised by *EFY* Group. She explains how communication standards that started off with non-IoT (Internet of Things) applications are now evolving to help the IoT.



The Internet of Things (IoT) thrives on communication standards, and these standards are evolving rapidly to support the IoT better. IoT applications range from home and building automation to smart cities, smart manufacturing, wearables, automotive and agriculture. Scope of the IoT is immense; however, depending on the use-case, IoT applications can pose different challenges for wireless communication standards.

Building a wireless communication standard for the IoT is hard with all the available communication standards. It is not easy to decide which one is best suited for a particular IoT application. There are standards spanning from personal area networks (PANs)

and proximity sensors to wide-area networks (WANs) and low-powered wide-area networks (LPWANs) that are specifically meant for the IoT. And in between, there are also wireless LANs (WLANs). Different communication standards offer different specifications to help the IoT in different ways. These communication standards are evolving and enabling new and future IoT applications.

The IoT drives communication standards on both (opposite) ends. It needs low latency (fast response time) as well as low power. Some applications require high data rate and wide range. These are conflicting goals and are difficult for a single wireless standard to achieve. On top of this, there are other things to be resolved such as

security, low cost, network capacity and reliability. This is the real challenge for wireless communication standards.

Evolving wireless standards in the IoT communication

Some wireless standards that are evolving to help the IoT are 802.11 for WLANs (802.11ax and 802.11ba that are coming up as new standards), Bluetooth 5, sub-1GHz and narrow-band IoT (NB-IoT), which is meant specifically for the IoT.

802.11ba is also called wake-up receiver. Usually, WLAN is used for applications that require very high data rate and where power is not a concern. But to enable the IoT, WLAN has to reduce its power consumption. The new standard of WLAN, 802.11ba, works

